

TE Curve Verification Pipeline Selected Active Model Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates only the currently selected active model families against the repository held-out TE-curve test split. The `global` surface is intentionally excluded from this reduced decision report.

Dataset And Split

- dataset config: `config/datasets/transmission_error_dataset.yaml` ;
- dataset root: `data\simplified_dataset` ;
- comparison mode: `selected_active_model_matrix` ;
- candidate count: `7` ;
- held-out curve count before candidate filtering: `97` ;
- percentage-error denominator: `peak_to_peak_truth` ;
- evaluated surface: `Bw` ;
- evaluated direction: `backward` ;

Candidate Inventory

Candidate	Source	Family	Surface	Valid Directions
<code>simplified_setpoints_feedforward_Bw</code>	<code>simplified_setpoints_model_archive</code>	<code>feedforward</code>	<code>Bw</code>	<code>backward</code>
<code>simplified_setpoints_tree_Bw</code>	<code>simplified_setpoints_model_archive</code>	<code>tree</code>	<code>Bw</code>	<code>backward</code>
<code>simplified_setpoints_harmonic_regression_Bw</code>	<code>simplified_setpoints_model_archive</code>	<code>harmonic_regression</code>	<code>Bw</code>	<code>backward</code>
<code>simplified_setpoints_periodic_mlp_harmonic_Bw</code>	<code>simplified_setpoints_model_archive</code>	<code>periodic_mlp_harmonic</code>	<code>Bw</code>	<code>backward</code>
<code>simplified_setpoints_periodic_gru_sequence_Bw</code>	<code>simplified_setpoints_model_archive</code>	<code>periodic_gru_sequence</code>	<code>Bw</code>	<code>backward</code>
<code>simplified_setpoints_wave4_1_mae_robust_loss_Bw</code>	<code>simplified_setpoints_model_archive</code>	<code>wave4_1_mae_robust_loss</code>	<code>Bw</code>	<code>backward</code>
<code>simplified_setpoints_wave4_2_quantile_p10_p50_p90_Bw</code>	<code>simplified_setpoints_model_archive</code>	<code>wave4_2_quantile_p10_p50_p90</code>	<code>Bw</code>	<code>backward</code>

Exact Model Paths

Candidate	Family	Surface	ONNX Model Path	Python Model Path
simplified_setpoints_feedforward_Bw	feedforward	Bw	models/simplified_dataset/setpoints/exported/feedforward/backward/2026-07-07-16-34-46_te_feedforward_bw_simplified_setpoints/onnx/model.onnx	models/simplified_dataset/setpoints/exported/feedforward/backward/2026-07-07-16-34-46_te_feedforward_bw_simplified_setpoints/python/feedforward-epoch=128-val_mae=0.00297364.ckpt
simplified_setpoints_tree_Bw	tree	Bw	models/simplified_dataset/setpoints/exported/tree/backward/2026-07-07-02-33-43_te_tree_bw_simplified_setpoints/onnx/model.onnx	models/simplified_dataset/setpoints/exported/tree/backward/2026-07-07-02-33-43_te_tree_bw_simplified_setpoints/python/tree_model.pkl
simplified_setpoints_harmonic_regression_Bw	harmonic_regression	Bw	models/simplified_dataset/setpoints/exported/harmonic_regression/backward/2026-07-07-23-01-08_te_harmonic_regression_bw_simplified_setpoints/onnx/model.onnx	models/simplified_dataset/setpoints/exported/harmonic_regression/backward/2026-07-07-23-01-08_te_harmonic_regression_bw_simplified_setpoints/python/harmonic_regression-epoch=060-val_mae=0.01698885.ckpt
simplified_setpoints_periodic_mlp_harmonic_Bw	periodic_mlp_harmonic	Bw	models/simplified_dataset/setpoints/exported/periodic_mlp_harmonic/backward/2026-07-08-01-19-22_te_periodic_mlp_harmonic_bw_simplified_setpoints/onnx/model.onnx	models/simplified_dataset/setpoints/exported/periodic_mlp_harmonic/backward/2026-07-08-01-19-22_te_periodic_mlp_harmonic_bw_simplified_setpoints/python/periodic_mlp-epoch=053-val_mae=0.00280310.ckpt
simplified_setpoints_periodic_gru_sequence_Bw	periodic_gru_sequence	Bw	models/simplified_dataset/setpoints/exported/periodic_gru_sequence/backward/2026-07-08-22-23-26_te_periodic_gru_sequence_bw_simplified_setpoints/onnx/model.onnx	models/simplified_dataset/setpoints/exported/periodic_gru_sequence/backward/2026-07-08-22-23-26_te_periodic_gru_sequence_bw_simplified_setpoints/python/periodic_gru_sequence-epoch=081-val_mae=0.00349987.ckpt
simplified_setpoints_wave4_1_mae_robust_loss_Bw	wave4_1_mae_robust_loss	Bw	models/simplified_dataset/setpoints/exported/wave4_1_mae_robust_loss/backward/2026-07-12-14-58-28_te_wave4_1_mae_robust_loss_bw_simplified_setpoints/onnx/model.onnx	models/simplified_dataset/setpoints/exported/wave4_1_mae_robust_loss/backward/2026-07-12-14-58-28_te_wave4_1_mae_robust_loss_bw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=073-val_mae=0.00358581.ckpt
simplified_setpoints_wave4_2_quantile_p10_p50_p90_Bw	wave4_2_quantile_p10_p50_p90	Bw	models/simplified_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-11-58-31_te_wave4_2_quantile_p10_p50_p90_bw_simplified_setpoints/onnx/model.onnx	models/simplified_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-11-58-31_te_wave4_2_quantile_p10_p50_p90_bw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=087-val_mae=0.00355108.ckpt

Metric Ranking

Rank	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
1	simplified_setpoints_tree_Bw	0.003345	0.003754	7.249	13.745
2	simplified_setpoints_periodic_gru_sequence_Bw	0.003364	0.003671	7.402	13.429
3	simplified_setpoints_wave4_2_quantile_p10_p50_p90_Bw	0.003473	0.003784	7.598	14.047
4	simplified_setpoints_wave4_1_mae_robust_loss_Bw	0.003519	0.003851	7.680	12.958
5	simplified_setpoints_feedforward_Bw	0.003594	0.004023	7.858	15.119
6	simplified_setpoints_periodic_mlp_harmonic_Bw	0.003842	0.004144	8.477	16.308
7	simplified_setpoints_harmonic_regression_Bw	0.017958	0.018234	41.314	85.556

Direction Breakdown

Direction	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
backward	simplified_setpoints_tree_Bw	0.003345	0.003754	7.249	13.745
backward	simplified_setpoints_periodic_gru_sequence_Bw	0.003364	0.003671	7.402	13.429
backward	simplified_setpoints_wave4_2_quantile_p10_p50_p90_Bw	0.003473	0.003784	7.598	14.047
backward	simplified_setpoints_wave4_1_mae_robust_loss_Bw	0.003519	0.003851	7.680	12.958
backward	simplified_setpoints_feedforward_Bw	0.003594	0.004023	7.858	15.119
backward	simplified_setpoints_periodic_mlp_harmonic_Bw	0.003842	0.004144	8.477	16.308
backward	simplified_setpoints_harmonic_regression_Bw	0.017958	0.018234	41.314	85.556

Curve Evidence

Each selected candidate is shown with the same four deterministic held-out operating conditions for this direction. The dark line is the measured TE curve and the blue line is the model prediction for the same operating condition.

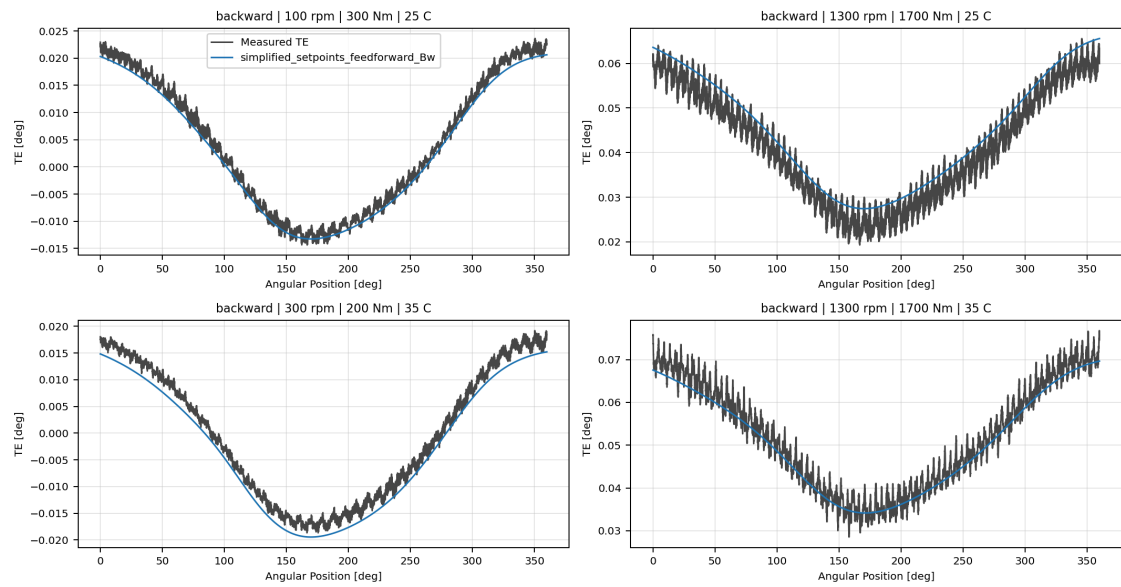
Shared operating conditions

- 100 rpm, 300 Nm, 25 C
- 1300 rpm, 1700 Nm, 25 C
- 300 rpm, 200 Nm, 35 C
- 1300 rpm, 1700 Nm, 35 C

Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
simplified_setpoints_feedforward_Bw	Selected candidate	0.003594	0.004023	7.858	15.119
simplified_setpoints_tree_Bw	Surface leader	0.003345	0.003754	7.249	13.745
simplified_setpoints_harmonic_regression_Bw	Selected candidate	0.017958	0.018234	41.314	85.556
simplified_setpoints_periodic_mlp_harmonic_Bw	Selected candidate	0.003842	0.004144	8.477	16.308
simplified_setpoints_periodic_gru_sequence_Bw	Selected candidate	0.003364	0.003671	7.402	13.429
simplified_setpoints_wave4_1_mae_robust_loss_Bw	Selected candidate	0.003519	0.003851	7.680	12.958
simplified_setpoints_wave4_2_quantile_p10_p50_p90_Bw	Selected candidate	0.003473	0.003784	7.598	14.047

simplified_setpoints_feedforward_Bw

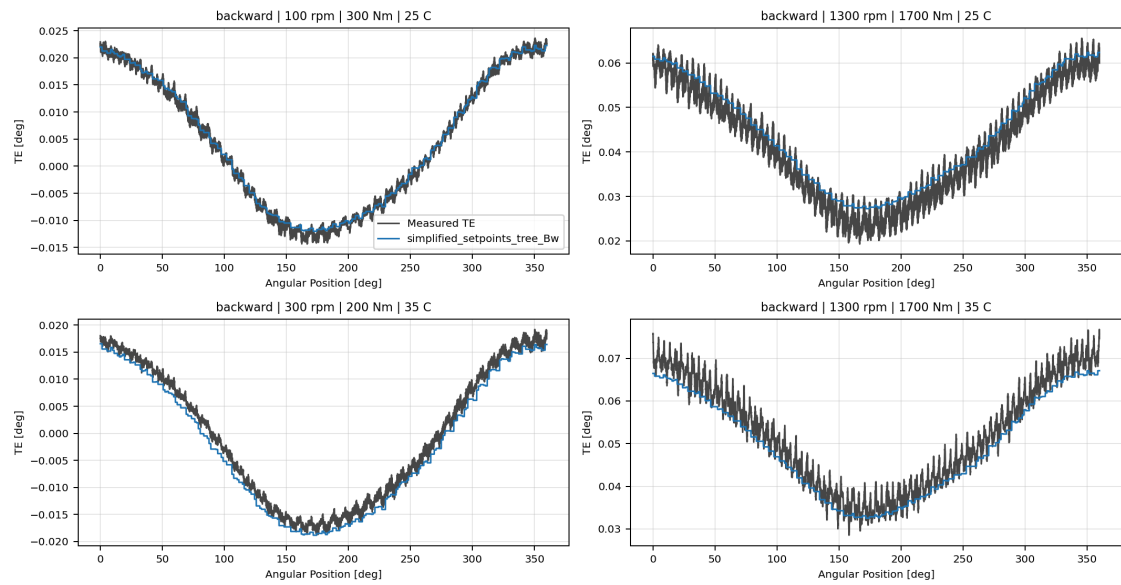
simplified_setpoints_feedforward_Bw



simplified_setpoints_feedforward_Bw measured-versus-predicted TE collage

simplified_setpoints_tree_Bw

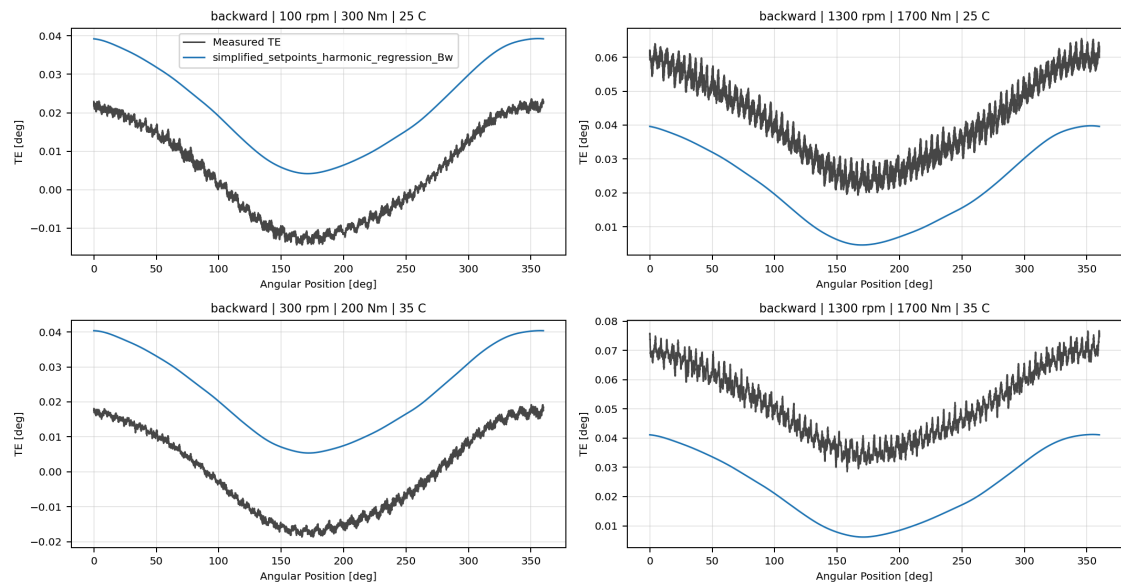
simplified_setpoints_tree_Bw



simplified_setpoints_tree_Bw measured-versus-predicted TE collage

simplified_setpoints_harmonic_regression_Bw

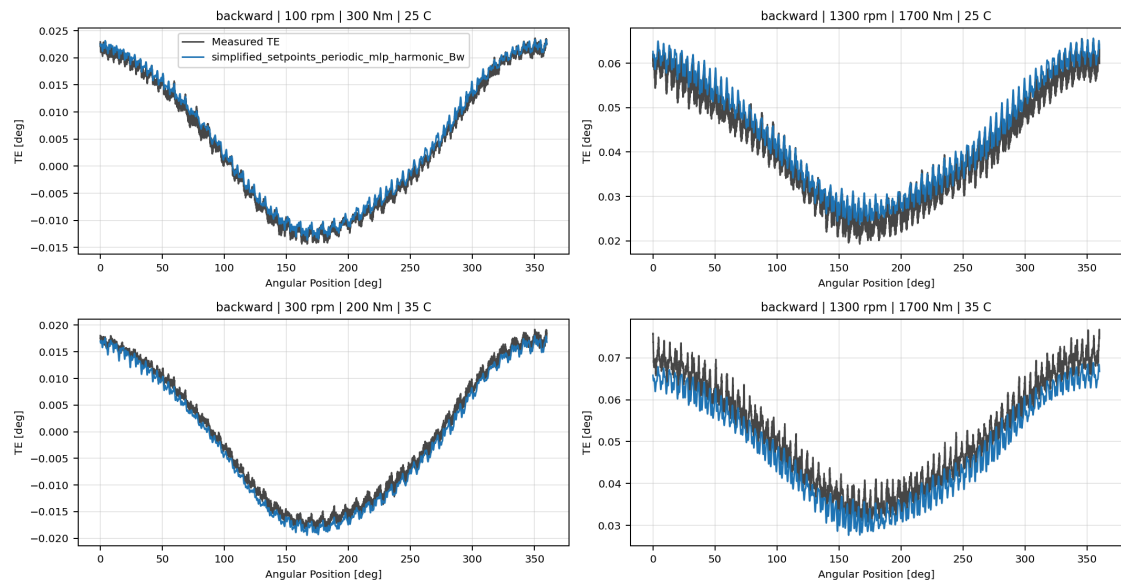
simplified_setpoints_harmonic_regression_Bw



simplified_setpoints_harmonic_regression_Bw measured-versus-predicted TE collage

simplified_setpoints_periodic_mlp_harmonic_Bw

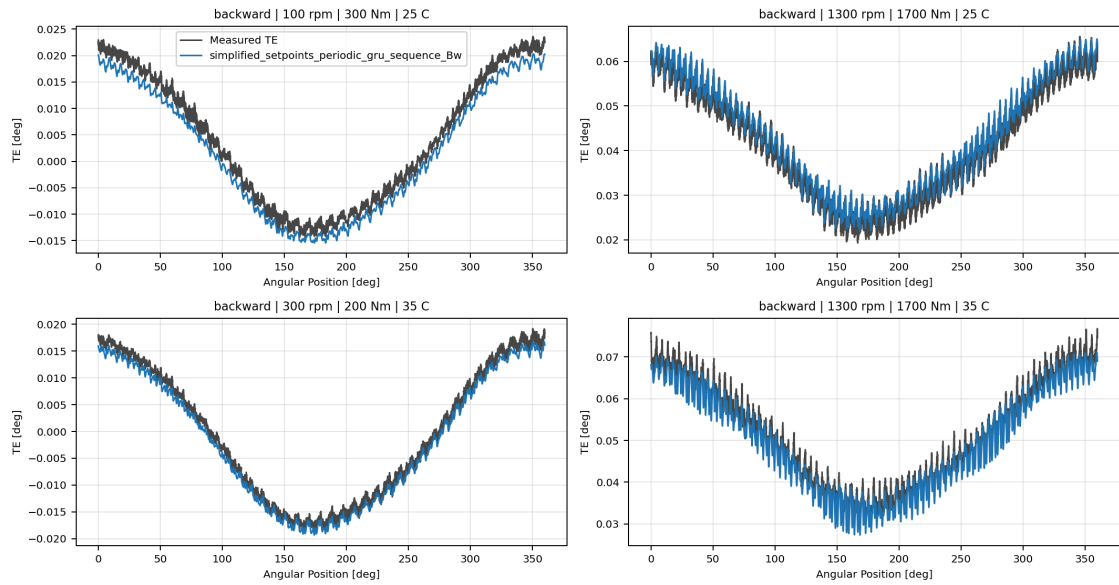
simplified_setpoints_periodic_mlp_harmonic_Bw



simplified_setpoints_periodic_mlp_harmonic_Bw measured-versus-predicted TE collage

simplified_setpoints_periodic_gru_sequence_Bw

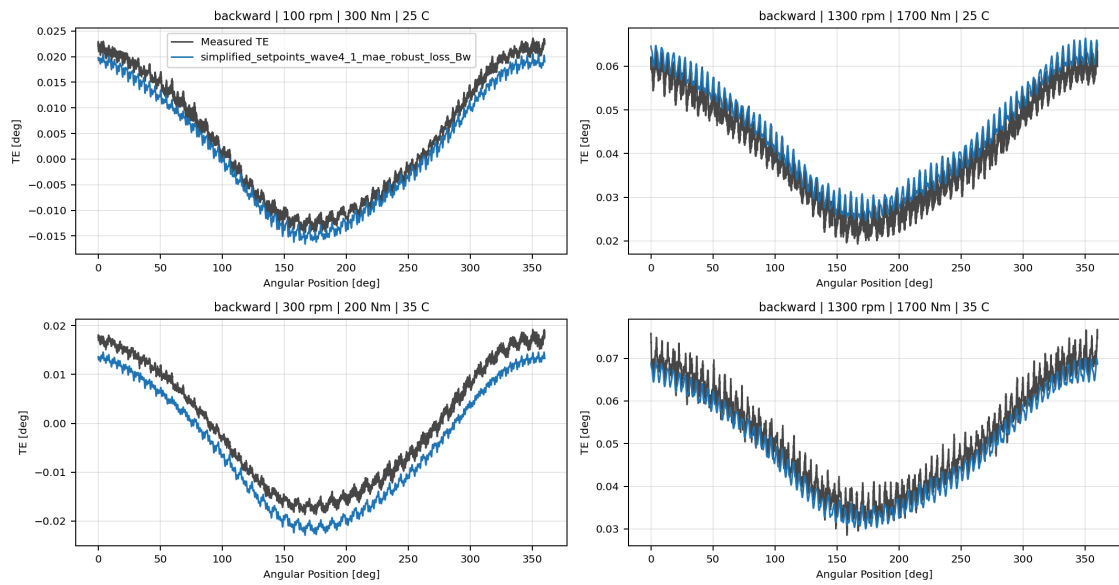
simplified_setpoints_periodic_gru_sequence_Bw



simplified_setpoints_periodic_gru_sequence_Bw measured-versus-predicted TE collage

simplified_setpoints_wave4_1_mae_robust_loss_Bw

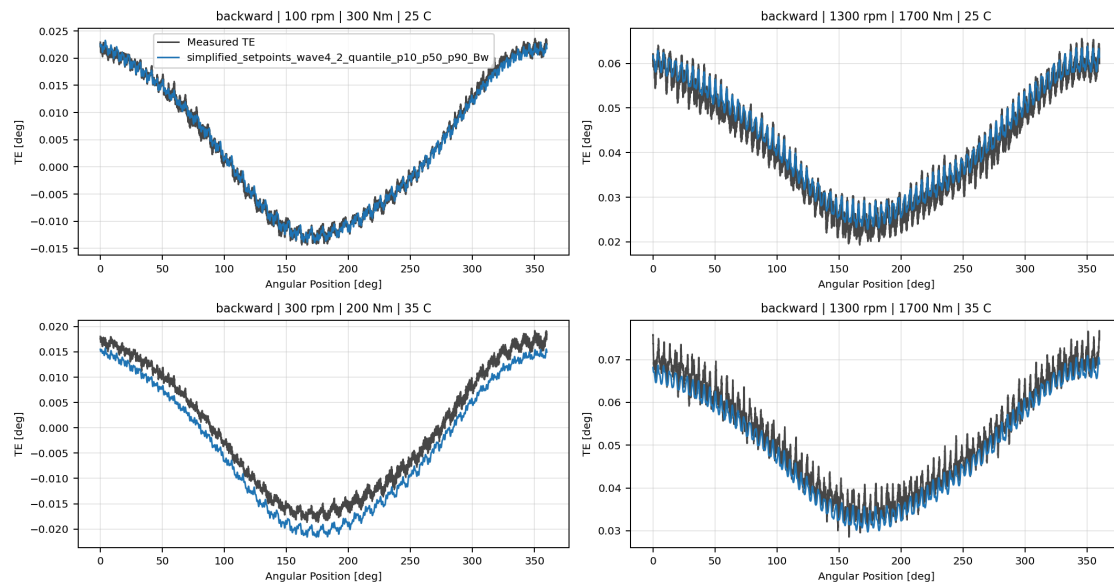
simplified_setpoints_wave4_1_mae_robust_loss_Bw



simplified_setpoints_wave4_1_mae_robust_loss_Bw measured-versus-predicted TE collage

simplified_setpoints_wave4_2_quantile_p10_p50_p90_Bw

simplified_setpoints_wave4_2_quantile_p10_p50_p90_Bw



simplified_setpoints_wave4_2_quantile_p10_p50_p90_Bw measured-versus-predicted TE collage

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-19-17-32-44_track2_selected_active_simplified_setpoints_matrix_track2_selected_active_simplified_setpoints_backward_2026_07_19\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-19-17-32-44_track2_selected_active_simplified_setpoints_matrix_track2_selected_active_simplified_setpoints_backward_2026_07_19\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots\selected_active_model_matrix` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within this selected-active surface report. The curve-evidence section uses shared deterministic held-out operating conditions so visual shape fidelity can be compared across the selected families.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- RCIM Model-Bank Reproduction remains a separate paper-reference benchmark path and is not part of this selected-active model-only report.